



Topic Test: OxfordAQA
International A level Chemistry
A2 Organic Chemistry

Name: _____

Class: _____

Date: _____

Time: **80 minutes**

Marks: **76 marks**

Comments:

1

This question is about three isomers of C_4H_8O

A	B	C
$CH_3CH_2COCH_3$	$CH_3CH_2CH_2CHO$	$CH_3CH=CHCH_2OH$

- (a) Give the IUPAC name for isomer **B** and for isomer **C**.

B _____

C _____

(2)

- (b) Isomers **A** and **B** both react with KCN followed by dilute acid.

Name the mechanism for this reaction.

Draw the two enantiomers formed by isomer **A** in this reaction and show how the two enantiomers are related to each other.

Mechanism _____

Enantiomer 1

Enantiomer 2

(3)

(c) Describe a test-tube reaction to distinguish isomer **C** from isomer **A**.

State the observations.

Test _____

Observation(s) with isomer **A** _____

Observation(s) with isomer **C** _____

(2)

(Total 7 marks)

2

Esters have many important commercial uses such as solvents and artificial flavourings in foods.

Esters can be prepared in several ways including the reactions of alcohols with carboxylic acids, acid anhydrides, acyl chlorides and other esters.

(a) Ethyl butanoate is used as a pineapple flavouring in sweets and cakes.

Write an equation for the preparation of ethyl butanoate from an acid and an alcohol.

Give a catalyst used for the reaction.

(4)

- (b) Butyl ethanoate is used as a solvent in the pharmaceutical industry.

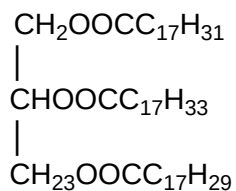
Write an equation for the preparation of butyl ethanoate from an acid anhydride and an alcohol.

(3)

- (c) Name and outline a mechanism for the reaction of CH_3COCl with CH_3OH to form an ester.

(5)

- (d) The ester shown below occurs in vegetable oils. Write an equation to show the formation of biodiesel from this ester.



(3)

- (e) Draw the repeating unit of the polyester Terylene that is made from benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

Although Terylene is biodegradeable, it is preferable to recycle objects made from Terylene.

Give **one** advantage and **one** disadvantage of recycling objects made from Terylene.

(4)

(Total 19 marks)

3

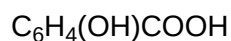
Describe briefly how you could measure the melting point of aspirin.

(Total 2 marks)

4

Salicylic acid, $\text{C}_6\text{H}_4(\text{OH})\text{COOH}$, reacts with magnesium to produce magnesium salicylate and hydrogen.

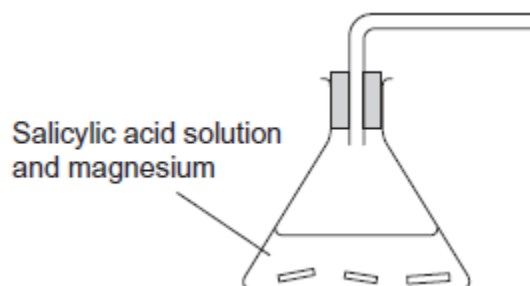
(a) Complete the equation for this reaction.



(1)

(b) In an alternative method for determining percentage purity, a student reacted a solution of salicylic acid with an excess of magnesium and collected the hydrogen gas that was released.

Complete the diagram below to show an apparatus that could be used to collect and measure the volume of hydrogen gas produced.

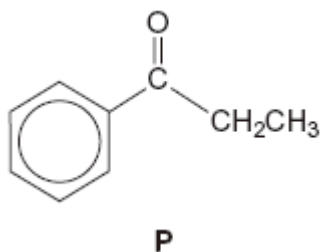


(1)

(Total 2 marks)

5

Consider compound **P** shown below that is formed by the reaction of benzene with an electrophile.



- (a) Give the **two** substances that react together to form the electrophile and write an equation to show the formation of this electrophile.

(3)

- (b) Outline a mechanism for the reaction of this electrophile with benzene to form **P**.

(3)

- (c) Compound **Q** is an isomer of **P** that shows optical isomerism. **Q** forms a silver mirror when added to a suitable reagent.

Identify this reagent and suggest a structure for **Q**.

(2)

(Total 8 marks)

6

Ammonia and ethylamine are examples of weak Brønsted–Lowry bases.

- (a) State the meaning of the term *Brønsted–Lowry base*.

(1)

- (b) (i) Write an equation for the reaction of ethylamine ($\text{CH}_3\text{CH}_2\text{NH}_2$) with water to form a weakly alkaline solution.

(1)

- (ii) In terms of this reaction, state why the solution formed is **weakly** alkaline.

(1)

- (c) State which is the stronger base, ammonia or ethylamine. Explain your answer.

Stronger base _____

Explanation _____

(3)

- (d) Give the formula of an organic compound that forms an alkaline buffer solution when added to a solution of ethylamine.

(1)

- (e) Explain qualitatively how the buffer solution in part (d) maintains an almost constant pH when a small amount of hydrochloric acid is added to it.

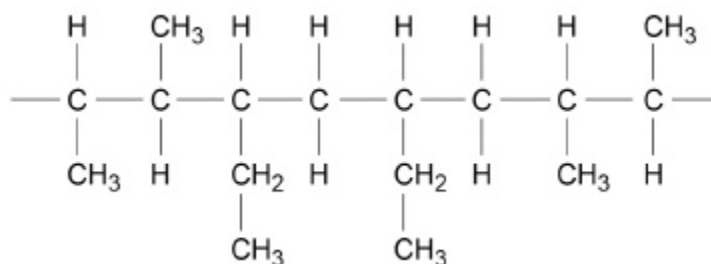
(2)

(Total 9 marks)

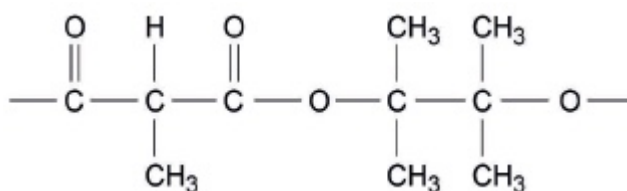
7

Sections of two polymers, **E** and **F**, are shown.

E



F



- (a) Polymer **E** is made from two different alkene monomers.

Draw the skeletal formula of each alkene monomer.

Alkene monomer 1

Alkene monomer 2

(2)

- (b) Polymer **F** is a polyester. This polyester is produced by reacting a diol with a dicarboxylic acid.

Give the structure of the diol and the structure of the dicarboxylic acid used to produce this polyester.

Diol

Dicarboxylic acid

(2)

- (c) Suggest an environmental advantage of polymer **F** compared with polymer **E**.

Explain your answer.

Advantage _____

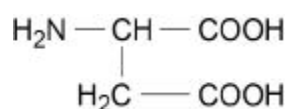
Explanation _____

(3)

(Total 7 marks)

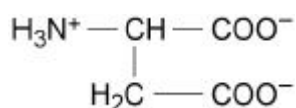
8

The structure of aspartic acid is

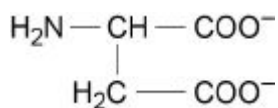


Which structure is formed by aspartic acid in solution at pH 14?

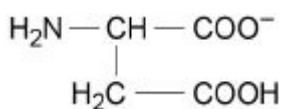
A


☐

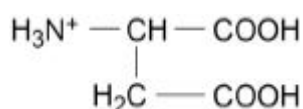
B


☐

C


☐

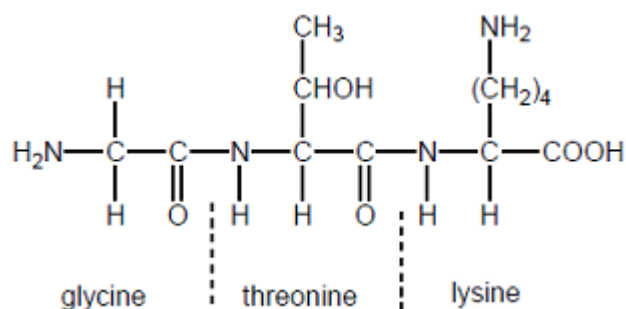
D


☐

(Total 1 mark)

9

The tripeptide shown in the following figure is formed from the amino acids glycine, threonine and lysine.



- (a) Draw a separate circle around **each** of the asymmetric carbon atoms in the tripeptide in the figure.

(1)

- (b) Draw the zwitterion of glycine.

(1)

- (c) Draw the structure of the species formed when glycine reacts with an excess of bromomethane.

(1)

- (d) Deduce the IUPAC name of threonine.

(1)

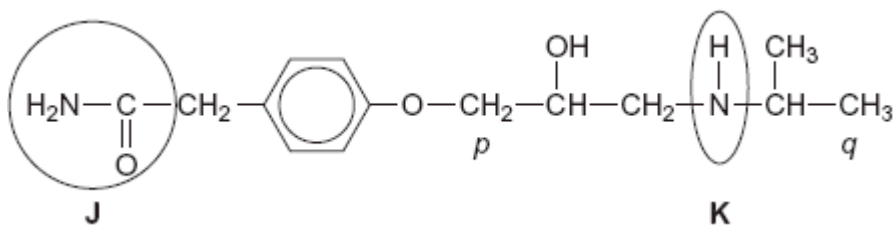
- (e) Draw the structure of the species formed by lysine at low pH.

(1)

(Total 5 marks)

10

Atenolol is an example of the type of medicine called a beta blocker. These medicines are used to lower blood pressure by slowing the heart rate. The structure of atenolol is shown below.



- (a) Give the name of each of the circled functional groups labelled **J** and **K** on the structure of atenolol shown above.

Functional group labelled **J** _____

Functional group labelled **K** _____

(2)

- (b) The ^1H n.m.r. spectrum of atenolol was recorded.

One of the peaks in the ^1H n.m.r. spectrum is produced by the CH_2 group labelled *p* in the structure of atenolol.

Use **Table 2** on the Data Sheet to suggest a range of δ values for this peak.

Name the splitting pattern of this peak.

Range of δ values _____

Name of splitting pattern _____

(2)

- (c) N.m.r. spectra are recorded using samples in solution.

The ^1H n.m.r. spectrum was recorded using a solution of atenolol in CDCl_3

- (i) Suggest why CDCl_3 and **not** CHCl_3 was used as the solvent.

(1)

- (ii) Suggest why CDCl_3 is a more effective solvent than CCl_4 for polar molecules such as atenolol.

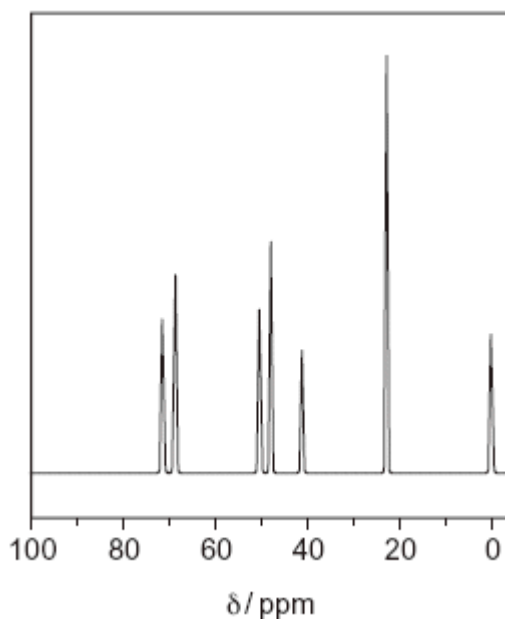
(1)

- (d) The ^{13}C n.m.r. spectrum of atenolol was also recorded.

Use the structure of atenolol given to deduce the total number of peaks in the ^{13}C n.m.r. spectrum of atenolol.

(1)

- (e) Part of the ^{13}C n.m.r. spectrum of atenolol is shown below. Use this spectrum and **Table 3** on the Data Sheet, where appropriate, to answer the questions which follow.



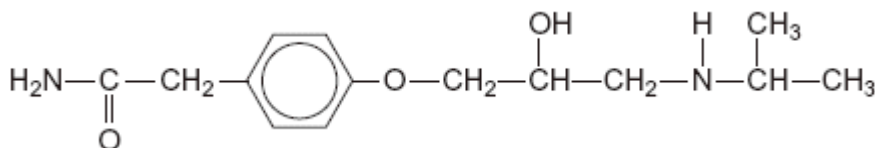
- (i) Give the formula of the compound that is used as a standard and produces the peak at $\delta = 0$ ppm in the spectrum.

(1)

- (ii) One of the peaks in the ^{13}C n.m.r. spectrum above is produced by the CH_3 group labelled *q* in the structure of atenolol. Identify this peak in the spectrum by stating its δ value.

(1)

- (iii) There are three CH_2 groups in the structure of atenolol. One of these CH_2 groups produces the peak at $\delta = 71$ in the ^{13}C n.m.r. spectrum above. Draw a circle around this CH_2 group in the structure of atenolol shown below.



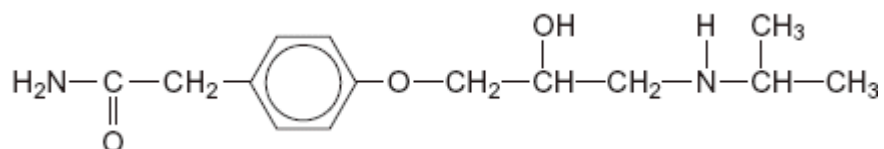
(1)

- (f) Atenolol is produced industrially as a racemate (an equimolar mixture of two enantiomers) by reduction of a ketone. Both enantiomers are able to lower blood pressure. However, recent research has shown that one enantiomer is preferred in medicines.

(i) Suggest a reducing agent that could reduce a ketone to form atenolol.

(1)

(ii) Draw a circle around the asymmetric carbon atom in the structure of atenolol shown below.



(1)

(iii) Suggest how you could show that the atenolol produced by reduction of a ketone was a racemate and **not** a single enantiomer.

(2)

(iv) Suggest **one** advantage and **one** disadvantage of using a racemate rather than a single enantiomer in medicines.

Advantage _____

Disadvantage _____

(2)

(Total 16 marks)

Mark schemes

1

(a) (B) Butanal

1

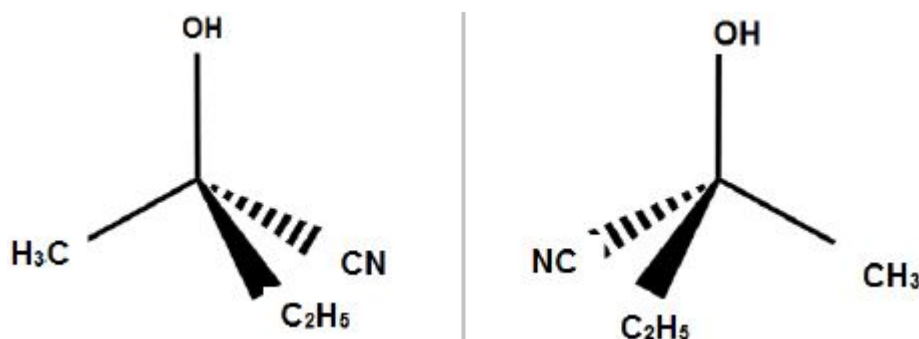
(C) But-2-en-1-ol

1

(b) (Mechanism) Nucleophilic addition

1

(Enantiomers) or 2D equivalent mirror images.



2

(c) (Test) Add bromine water

Allow addition of acidified / H^+ potassium dichromate(VI)

1

(Observation) decolourises with C and no visible change with A

With potassium dichromate(VI)... (orange to) green with C and no visible change with A

1

[7]

2

(a) **M1** $CH_3CH_2CH_2COOH$
not C_3H_7COOH

1

M2 CH_3CH_2OH or C_2H_5OH

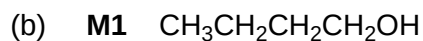
1

M3 $CH_3CH_2CH_2COOCH_2CH_3 + H_2O$
allow $C_3H_7COOC_2H_5$
penalise M3 for wrong products and unbalanced equation

1

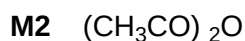
M4 H_2SO_4 or HCl or H_3PO_4 conc or dil or neither
not HNO_3

1



1

not $\text{C}_4\text{H}_9\text{OH}$



1

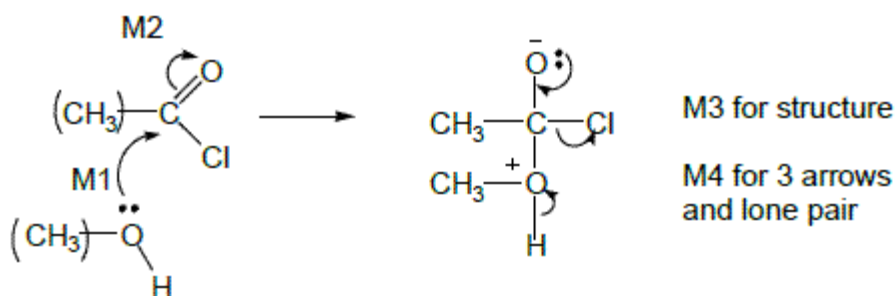


allow $\text{CH}_3\text{COOC}_4\text{H}_9$

penalise M3 for wrong products and unbalanced equation

1

(c) (nucleophilic) addition-elimination



not acylation alone

M2 not allowed indep of M1 but allow M1 for correct attack on $\text{C}^+=\text{O}$

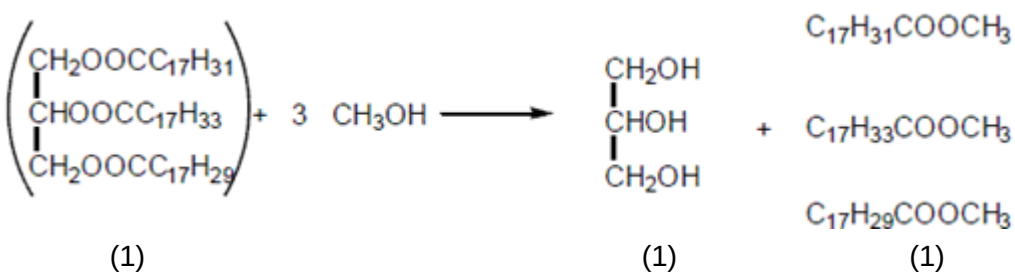
$\text{C}=\text{O}$ loses M2

only allow M4 after correct or v close M3

ignore Cl^- removing H^+

5

(d)



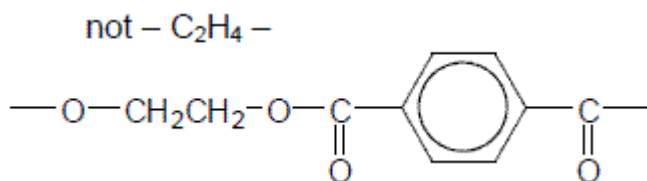
ignore errors in initial triester

First mark for $3\text{CH}_3\text{OH}$

Third mark for all three esters

3

(e)



First mark for correct ester link second mark for the rest including trailing bonds

If ester link wrong, lose second mark also

2

- Adv reduces landfill
 saves raw materials
 lower cost for recycling than making from scratch
 reduces CO₂ emissions by not being incinerated

not allow cost without qualification

ignore energy uses

1

- Disad difficulty/cost of collecting/sorting/processing
 product not suitable for original purpose, easily contaminated

not allow cost without qualification

ignore energy uses

1

[19]

3

Sample in capillary / melting point tube

Accept alternative as long as small container used

1

Heat in melting point apparatus / heat gently / slowly near melting point

1

[2]

4

(a) $\text{Mg} + 2\text{C}_6\text{H}_4(\text{OH})\text{COOH} \rightarrow (\text{C}_6\text{H}_4(\text{OH})\text{COO})_2\text{Mg} + \text{H}_2$

Accept multiples, including fractions.

1

(b) Gas syringe / inverted burette over water / measuring cylinder over water

*Collection apparatus must show graduations or be clearly labelled
(eg syringe, burette, measuring cylinder).*

1

[2]

5

- (a) $\text{CH}_3\text{CH}_2\text{COCl}$ OR $\text{CH}_3\text{CH}_2\text{CClO}$ OR propanoyl chloride
 OR $(\text{CH}_3\text{CH}_2\text{CO})_2\text{O}$ OR propanoic anhydride
 penalise contradiction in formula and name e.g. propyl chloride
could score in equation

1

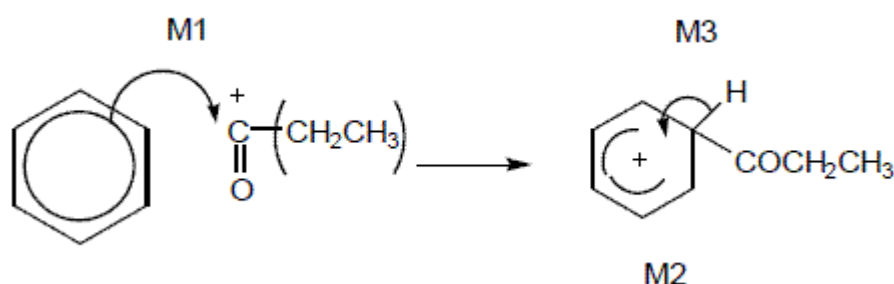
AlCl_3 or FeCl_3 or names
could score in equation

1

$\text{CH}_3\text{CH}_2\text{COCl} + \text{AlCl}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CO}^+ + \text{AlCl}_4^-$
 Allow RCOCl in equation but penalise above
allow + on C or O in equation

1

(b)

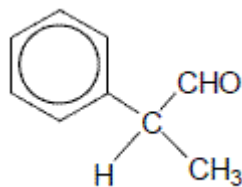


M1 arrow from circle or within it to C or to + on C
Horseshoe must not extend beyond C2 to C6 but can be smaller + not too close to C1
M3 arrow into hexagon unless Kekule
allow M3 arrow independent of M2 structure
Ignore base removing H in M3

3

- (c) Tollens or ammoniacal silver nitrate

1



penalise wrong formula

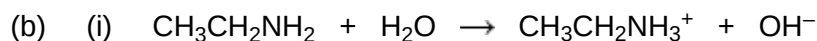
1

[8]

6

- (a) Proton acceptor

1



allow eq with or without $\rightleftharpoons$

allow $\text{C}_2\text{H}_5\text{NH}_2$ and $\text{C}_2\text{H}_5\text{NH}_3^+$ (plus can be on N or H or 3)

allow RHS as $\text{C}_2\text{H}_5\text{NH}_3\text{OH}$

1

(ii) Mark independently of (b)(i)

Allow

Ethylamine is only partly/slightly dissociated

OR

Ethylamine is only partly/slightly ionized

reaction/equilibrium lies to left or low $[\text{OH}^-]$ **OR** little OH^- formed

OR little ethylamine has reacted

Ignore "not fully dissociated" or "not fully ionized"

Ignore reference to ionisation or dissociation of water

1

(c) **M1** Ethylamine

If wrong no marks in (c)

1

M2 alkyl group is electron releasing/donating

OR alkyl group has (positive) inductive effect

1

M3 increases electron density on N(H₂)

OR increased availability of lp

OR increases ability of lp (to accept H^+)

Mark M3 is independent of M2

1

(d) $\text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$

Or any amine hydrochloride

allow name (ethylammonium chloride or ethylamine hydrochloride) or other halide for Cl

*or a strong **organic** acid*

NOT NH_4Cl

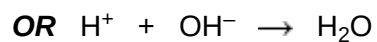
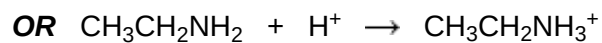
1

(e) Mark independently of (d)

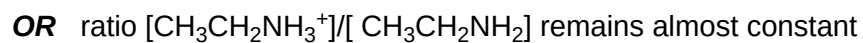
Extra H^+ reacts with ethylamine or OH^-

Or makes reference to Equilibrium (in (b)(i)) with amine on LHS

1



Equilibrium shifts to RHS



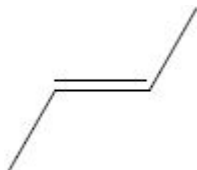
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[9]

7

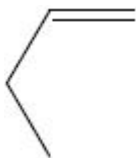
(a) Monomers:

Must be skeletal formulae



Allow 1 mark for two correct structural formulae.

1



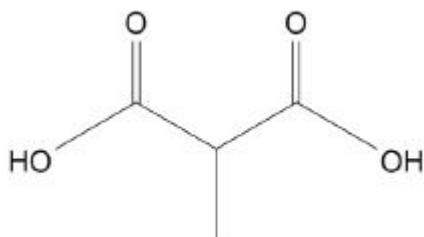
1

(b)



Allow structural formulae

1



Allow 1 mark for two correct structures the wrong way around.

1

- (c) Advantage: Polymer F is bio-degradeable
Allow F is easier to decompose.

1

Explanation:

Polar C=O group (in chain) in Polymer F

1

Allow converse arguments for E
Polymer E has strong C-C bonds

So F can be attacked by nucleophiles / water / can be hydrolysed

Polymer E cannot be attacked by nucleophiles / water /
cannot be hydrolysed

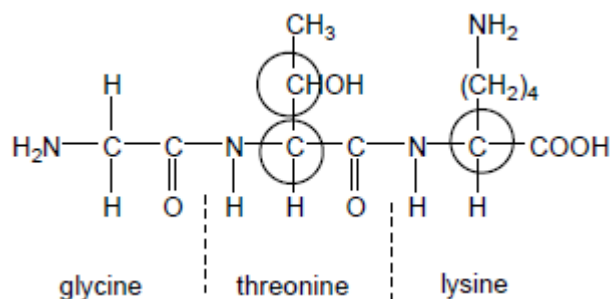
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[7]

8 B

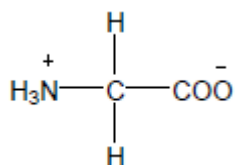
[1]

9 (a)



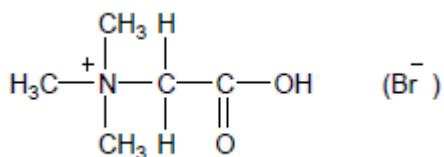
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(b)

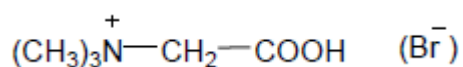


1

(c)



Allow

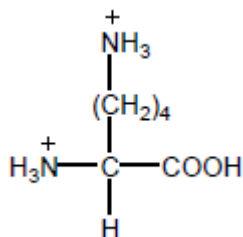


1

- (d) 2-amino-3-hydroxybutanoic acid

1

(e)



1

[5]

10

(a) J (acid) amide

not peptide, not N-substituted amide

1

K (secondary) amine or amino

*penalise primary or tertiary
allow N-substituted amine*

1

(b) (δ =) 3.1-3.9

1

doublet **OR** duplet

Not 3.7 – 4.1

Not secondary

name required not the number 2

1

(c) (i) Solvent must be proton-free

OR CHCl_3 has protons or has H or gives a peak

1

(ii) CDCl_3 is polar **OR** CCl_4 is non-polar

1

(d) 11 **OR** eleven

1

(e) (i) $\text{Si}(\text{CH}_3)_4$ **OR** $\text{SiC}_4\text{H}_{12}$

ignore TMS

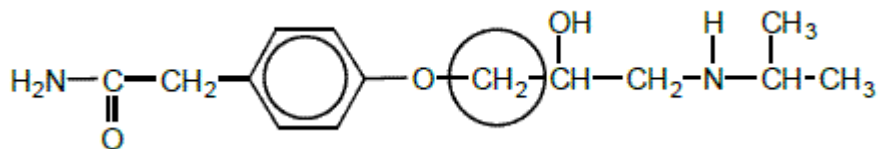
1

(ii) a single number or a range within 21-25

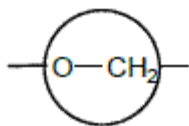
penalise anything outside this range

1

(iii)



allow ring around the C only and also allow



1

(f) (i) NaBH_4

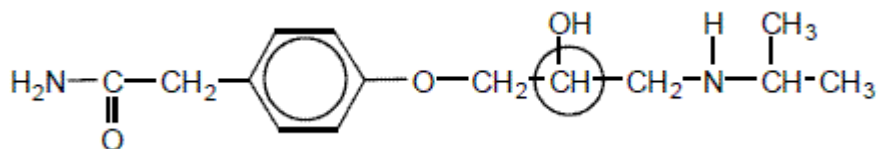
ignore name if formula correct

ignore solvent

allow LiAlH_4 Zn/HCl Sn/HCl H_2/Ni H_2/Pt

1

(ii)



allow ring around the C only

1

(iii) (plane) polarised light **OR** light in a polarimeter

1

polarised light is not rotated or is unaffected

penalise bent/diffracted/deflected/reflected

Not just solution is optically inactive

1

(iv) **adv** cheaper medicine due to cost or difficulty of separation or both can lower blood pressure

OR more effective/beneficial with a reason

or no need to separate

1

disadv may be side effects from one enantiomer in the mixture or only half the product works or one enantiomer may be ineffective or double dose required

1

[16]